XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 1: Units, Physical Quantities, and Vectors

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Physical Quantities

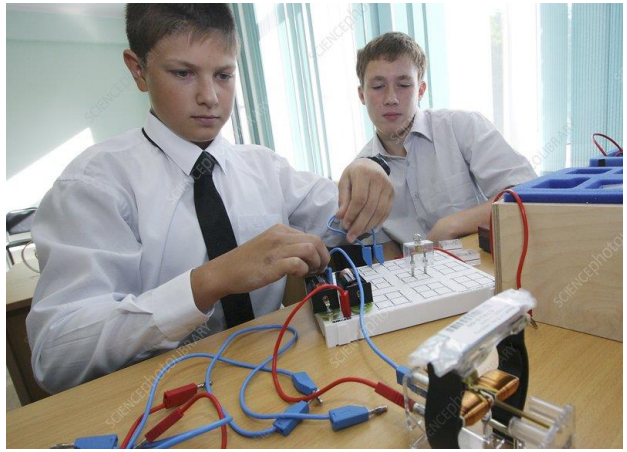
- Physics is one of the most fundamental of sciences, and physics is an experimental science.

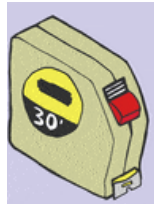


Physical Quantities

- Experiments require measurements.
- We generally use numbers to describe the results of measurements.
- Any number that is used to describe a physical phenomenon quantitatively is called a physical quantity.
- Examples of physical quantities in the engineering field

  - Electrical engineering: Voltage, Current, Electric Field Intensity
  - Mechanical engineering: Force, velocity, temperature, thermal fluid





Physical Quantities

- When measuring a physical quantity, one need to compare the results with reference standards.
- The International System (SI) of Units or metric system
  - The system of units as reference standards used by scientists and engineers around the world is commonly called “the metric system,” but since 1960 it has been known officially as the International System, or SI.

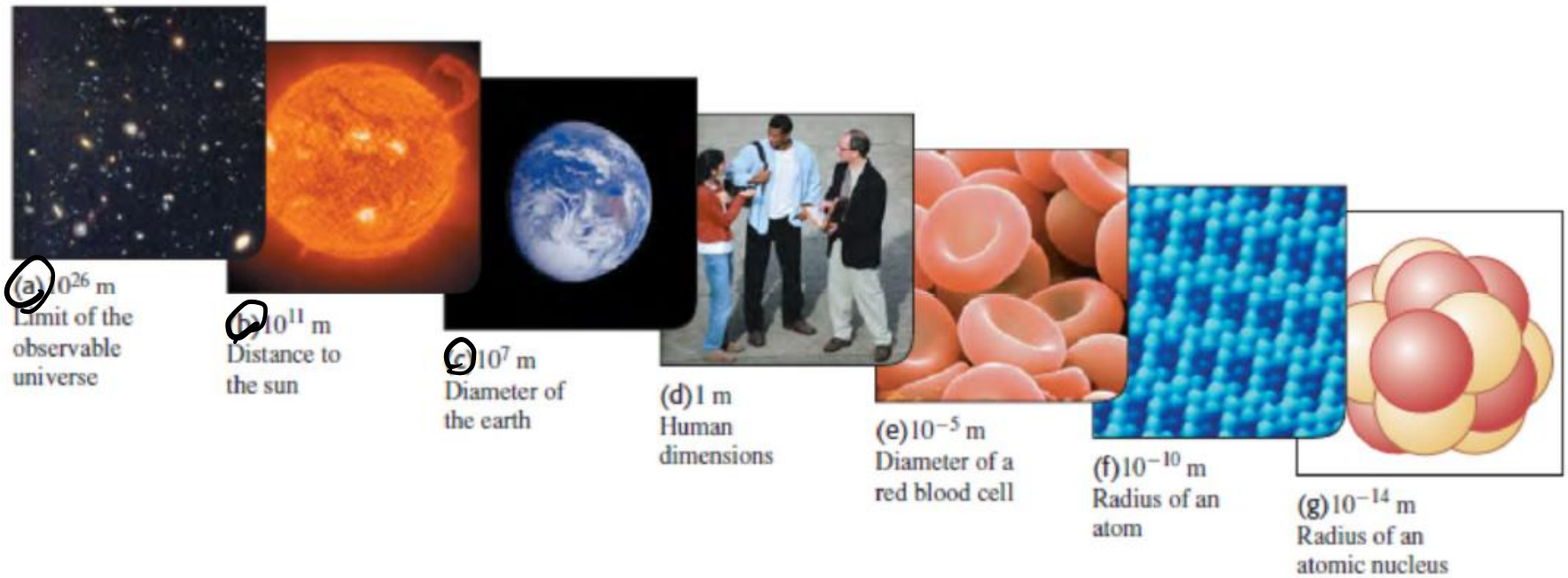


Seven Basic Quantities and Units

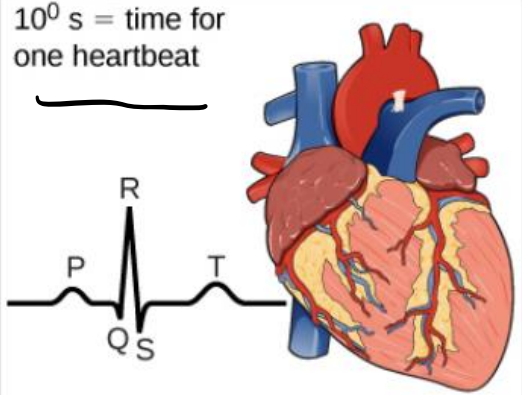
Quantity	Unit	Definition
Length	**m**	Distance light traveled in vacuum for 1/299792458 seconds
Mass	**kg**	Mass of a specific platinum-iridium alloy, for now
Time	**s**	9192631770 cycles of radiation of cesium-133
Current	A	That flows in 2 parallel wires resulting in force of 2×10^{-7} N / m on the wires.
Temperature	K	1/273.16 of thermodynamic temperature of triple point of water
Amount of substance	mol	That contains equal number of fundamental entities as 0.012 kg of carbon-12
Luminous intensity	cd	Of a source that emits monochromatic radiation of frequency 540×10^{12} Hz and that has a radiant intensity of 1/683 watt/steradian.

Typical Lengths in the Universe

- Physical sciences deal with the biggest, smallest, coldest, hottest, etc.
- Length also spans over many orders of magnitudes.



Other Quantities

Length in Meters (m)	Masses in Kilograms (kg)	Time in Seconds (s)
10^{-15} m = diameter of proton	10^{-30} kg = mass of electron	10^{-22} s = mean lifetime of very unstable nucleus
10^{-14} m = diameter of large nucleus	10^{-27} kg = mass of proton	10^{-17} s = time for single floating-point operation in a supercomputer
10^{-10} m = diameter of hydrogen atom	10^{-15} kg = mass of bacterium	10^{-15} s = time for one oscillation of visible light
10^{-7} m = diameter of typical virus	10^{-5} kg = mass of mosquito	10^{-13} s = time for one vibration of an atom in a solid
10^{-2} m = pinky fingernail width	10^{-2} kg = mass of hummingbird	10^{-3} s = duration of a nerve impulse
10^{0} m = height of 4 year old child 	10^{0} kg = mass of liter of water 	10^{0} s = time for one heartbeat 

Other Quantities

Length in Meters (m)	Masses in Kilograms (kg)	Time in Seconds (s)
10^2 m = length of football field	10^2 kg = mass of person	10^5 s = one day
10^7 m = diameter of Earth	10^{19} kg = mass of atmosphere	10^7 s = one year
10^{13} m = diameter of solar system	10^{22} kg = mass of Moon	10^9 s = human lifetime
10^{16} m = distance light travels in a year (one light-year)	10^{25} kg = mass of Earth	10^{11} s = recorded human history
10^{21} m = Milky Way diameter	10^{30} kg = mass of Sun	10^{17} s = age of Earth
10^{26} m = distance to edge of observable universe	10^{53} kg = upper limit on mass of known universe	10^{18} s = age of the universe

Unit Prefix and Notations

- Prefixes to some of the SI units simplify our work.
- Common ones used in Physics and Engineering:

- kilo – 10^3 (i.e., km, kW, kg)
- Mega – 10^6 (i.e., MHz, MΩ)
- Giga – 10^9 (i.e., GW, Gbyte)
- Terra – 10^{12} …

- deci – 10^{-1} (i.e., dB)
- centi – 10^{-2} (i.e., cm)
- milli – 10^{-3} (i.e., ml, mm)
- micro – 10^{-6} (i.e., μF)
- nano – 10^{-9} (i.e., nm)
- pico – 10^{-12}, femto – 10^{-15} …

no need to remember

Unit Prefix and Notations

Prefix	Symbol	Meaning	Prefix	Symbol	Meaning
yotta-	Y	10^{24}	yocto-	y	10^{-24}
zetta-	Z	10^{21}	zepto-	z	10^{-21}
exa-	E	10^{18}	atto-	a	10^{-18}
peta-	P	10^{15}	femto-	f	10^{-15}
tera-	T	10^{12}	pico-	p	10^{-12}
giga-	G	10^{9}	nano-	n	10^{-9}
mega-	M	10^{6}	micro-	μ	10^{-6}
kilo-	k	10^{3}	milli-	m	10^{-3}
hecto-	h	10^{2}	centi-	c	10^{-2}
deka-	da	10^{1}	deci-	d	10^{-1}

Metric Prefixes for Powers of 10 and Their Symbols

Unit Consistency

- Equations to express relationships among physical quantities, and they must always be **dimensionally consistent**.
- The British System or Imperial units.
  - Still widely used in country such as the United States and a few other countries. These Units of British Systems are being replaced by the Units of international Standards. For example:

Length: 1 inch = 2.54 cm (exactly)

Force: 1 pound = 4.448221615260 newtons (exactly)

will be given.

Unit Conversion

Length

$1\ \text{m} = 100\ \text{cm} = 1000\ \text{mm} = 10^6\ \mu\text{m} = 10^9\ \text{nm}$
$1\ \text{km} = 1000\ \text{m} = 0.6214\ \text{mi}$
$1\ \text{m} = 3.281\ \text{ft} = 39.37\ \text{in.}$
$1\ \text{cm} = 0.3937\ \text{in.}$
$1\ \text{in.} = 2.540\ \text{cm}$
$1\ \text{ft} = 30.48\ \text{cm}$
$1\ \text{yd} = 91.44\ \text{cm}$
$1\ \text{mi} = 5280\ \text{ft} = 1.609\ \text{km}$
$1\ \text{Å} = 10^{-10}\ \text{m} = 10^{-8}\ \text{cm} = 10^{-1}\ \text{nm}$
1 nautical mile $= 6080\ \text{ft}$
1 light-year $= 9.461 \times 10^{15}\ \text{m}$

Area

$1\ \text{cm}^2 = 0.155\ \text{in.}^2$
$1\ \text{m}^2 = 10^4\ \text{cm}^2 = 10.76\ \text{ft}^2$
$1\ \text{in.}^2 = 6.452\ \text{cm}^2$
$1\ \text{ft}^2 = 144\ \text{in.}^2 = 0.0929\ \text{m}^2$

Acceleration

$1\ \text{m/s}^2 = 100\ \text{cm/s}^2 = 3.281\ \text{ft/s}^2$
$1\ \text{cm/s}^2 = 0.01\ \text{m/s}^2 = 0.03281\ \text{ft/s}^2$
$1\ \text{ft/s}^2 = 0.3048\ \text{m/s}^2 = 30.48\ \text{cm/s}^2$
$1\ \text{mi/h} \cdot \text{s} = 1.467\ \text{ft/s}^2$

Mass

$1\ \text{kg} = 10^3\ \text{g} = 0.0685$ slug
$1\ \text{g} = 6.85 \times 10^{-5}$ slug
1 slug $= 14.59\ \text{kg}$
$1\ \text{u} = 1.661 \times 10^{-27}\ \text{kg}$
$1\ \text{kg}$ has a weight of $2.205\ \text{lb}$ when $g = 9.80\ \text{m/s}^2$

Force

$1\ \text{N} = 10^5\ \text{dyn} = 0.2248\ \text{lb}$
$1\ \text{lb} = 4.448\ \text{N} = 4.448 \times 10^5\ \text{dyn}$

Volume

1 liter = 1000 cm^3 = 10^{-3} m^3 = 0.03531 ft^3 = 61.02 $in.^3$
1 ft^3 = 0.02832 m^3 = 28.32 liters = 7.477 gallons
1 gallon = 3.788 liters

Time

1 min = 60 s
1 h = 3600 s
1 d = 86,400 s
1 y = 365.24 d = 3.156×10^7 s

Angle

1 rad = 57.30° = $180°/\pi$
1° = 0.01745 rad = $\pi/180$ rad
1 revolution = 360° = 2π rad
1 rev/min (rpm) = 0.1047 rad/s

Speed

1 m/s = 3.281 ft/s
1 ft/s = 0.3048 m/s
1 mi/min = 60 mi/h = 88 ft/s
1 km/h = 0.2778 m/s = 0.6214 mi/h
1 mi/h = 1.466 ft/s = 0.4470 m/s = 1.609 km/h
1 furlong/fortnight = 1.662×10^{-4} m/s

Pressure

1 Pa = 1 N/m^2 = 1.450×10^{-4} $lb/in.^2$ = 0.0209 lb/ft^2
1 bar = 10^5 Pa
1 $lb/in.^2$ = 6895 Pa
1 lb/ft^2 = 47.88 Pa
1 atm = 1.013×10^5 Pa = 1.013 bar
= 14.7 $lb/in.^2$ = 2117 lb/ft^2
1 mm Hg = 1 torr = 133.3 Pa

Energy

1 J = 10^7 ergs = 0.239 cal
1 cal = 4.186 J (based on 15° calorie)
1 ft · lb = 1.356 J
1 Btu = 1055 J = 252 cal = 778 ft · lb
1 eV = 1.602×10^{-19} J
1 kWh = 3.600×10^6 J

Mass–Energy Equivalence

1 kg $\leftrightarrow$ 8.988×10^{16} J
1 u $\leftrightarrow$ 931.5 MeV
1 eV $\leftrightarrow$ 1.074×10^{-9} u

Power

1 W = 1 J/s
1 hp = 746 W = 550 ft · lb/s
1 Btu/h = 0.293 W

Unit Conversions

Example 1.1 The world supersonic land speed record is 763.0 mi/h, set on October 15,1997, by Andy Green in the jet-engine car *Thrust SSC.* Express this speed in meters per second.

Solution:

To convert the units of a speed from mi/h to m/s, we can find these equalities from Appendix E of the textbook,

1 mi = 1.609 km, 1 km = 1000 m, 1 h = 3600 s.

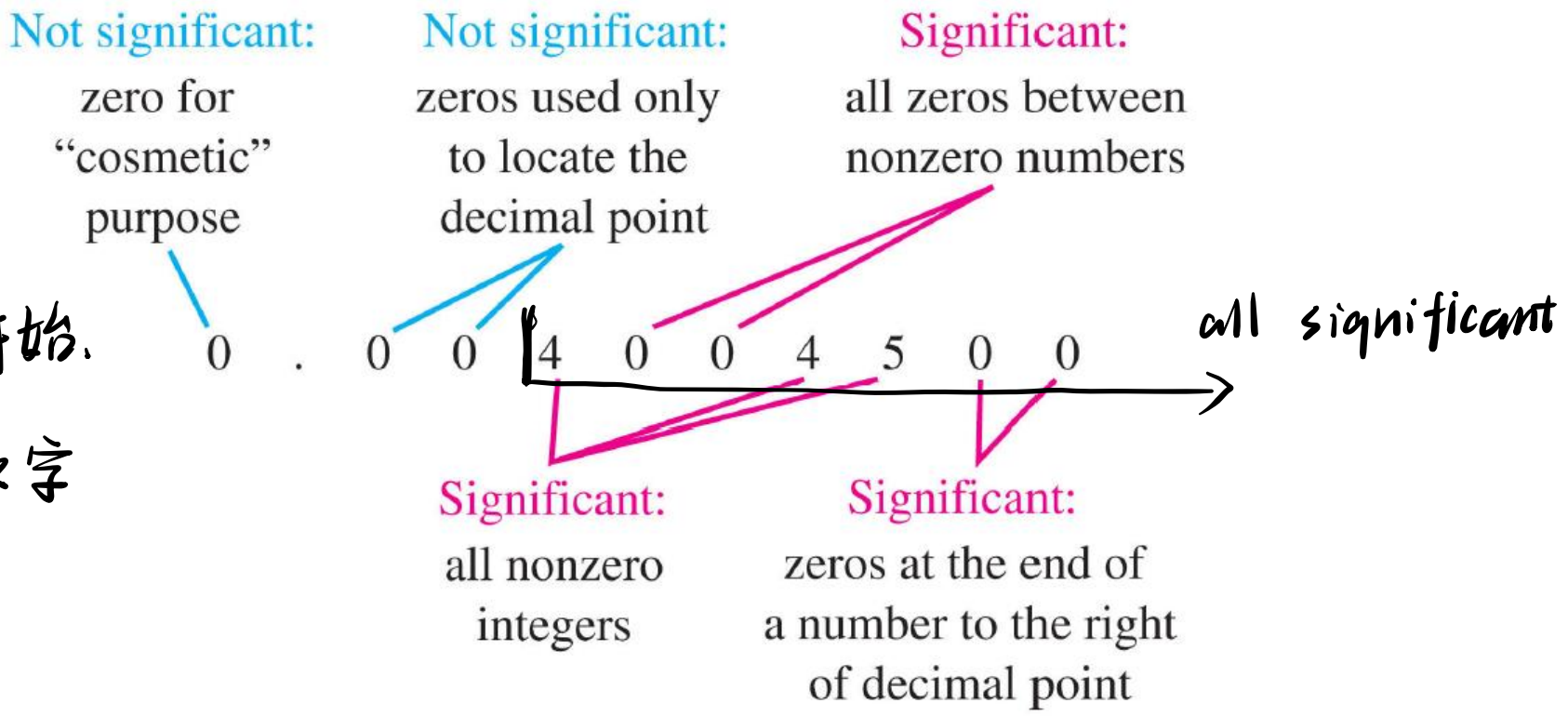
then we can get

$$763.0\ \text{mi/h} = \left(763.0\ \frac{\text{mi}}{\text{h}}\right)\left(\frac{1.609\ \text{km}}{1\ \text{mi}}\right)\left(\frac{1000\ \text{m}}{1\ \text{km}}\right)\left(\frac{1\ \text{h}}{3600\ \text{s}}\right)$$

$$= 341.0\ \text{m/s}$$

用表解决问题

Significant Figures



第1个非0数开始，都是有效数字

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Determining the number of significant figures in a quantity

Significant Figures

- A problem arises when a number ends with zeroes but contains no decimal point.
- In such cases, it is normally assumed that the zeroes are not significant.
- Scientific notation can be used to indicate whether end zeroes are significant.
- For example, a mass of 10,300 g can be written to show three, four, or five significant figures depending on how the measurement is obtained:

1.03×10^4 g (three significant figures)

1.030×10^4 g (four significant figures)

1.0300×10^4 g (five significant figures)

(The exponential term 10^4 does not add to the number of significant figures)

Significant Figures

- In our daily study, there is no strict rule for our calculation, but people have some custom or conventions:

  - to keep three significant figures after the decimal point
    0.857636 (round it off to) 0.858
    20.857636 (round it off to) 20.858

  - to keep three significant figures
    0.857636 (round it off to) 0.858
    20.857636 (round it off to) 20.9
    120.857636 (round it off to) 1.21×10^{2}
    (The significant figures start from the first non-zero figure)

more important

Significant Figures

- In scientific documents, be sure to follow the official requirements, i.e., the requirement is to keep **three** significant figures after the decimal point.

0.3 should be written as 0.300
0.32 should be written as 0.320
0.322 should be written as 0.322
0.3227 should be written as 0.323

Multiplication & Division Rule

- Answers are rounded to the number of digits that corresponds to the **least** number of significant figures in any of the numbers used in the calculation.

Result can have no more significant figures than the factor with the fewest significant figures:

$$\frac{0.745 \times 2.2}{3.885} = 0.42$$

$$1.32578 \times 10^{7} \times 4.11 \times 10^{-3} = 5.45 \times 10^{4}$$

Addition & Subtraction Rule

- Number of significant figures is determined by the term with the largest uncertainty (i.e., fewest digits to the right of the decimal point):

$$\begin{array}{rl} 3.6923 & 4 \\ +1.234 & 3 \\ +2.02 & 2 \\ \hline 6.9463 & \end{array}$$

round off to 6.95

$$\begin{array}{rl} 8.7937 & 4 \\ -2.123 & 3 \\ \hline 6.6707 & \end{array}$$

round off to 6.671

$$27.153 + 138.2 - 11.74 = 153.6$$

Significant Figures

- What is the best answer to report for (515 x 0.0025) + 24.57?

1.2875

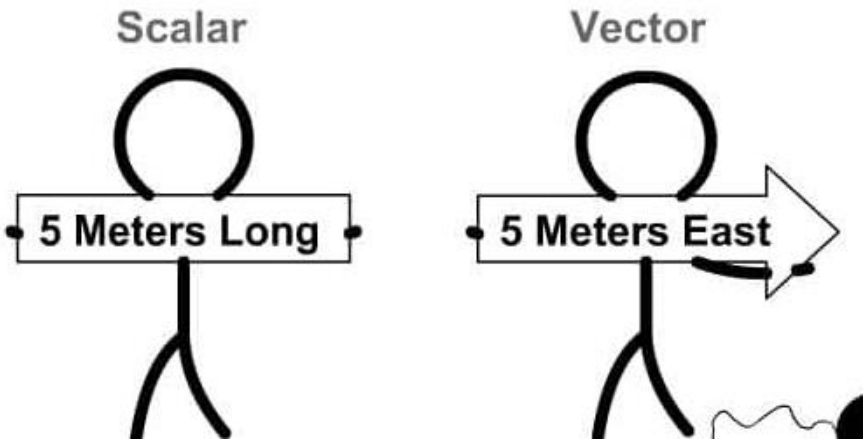
1.3 + 24.57 = 25.87

1 (1.3) 2 (24.57)

= 25.9 ✓

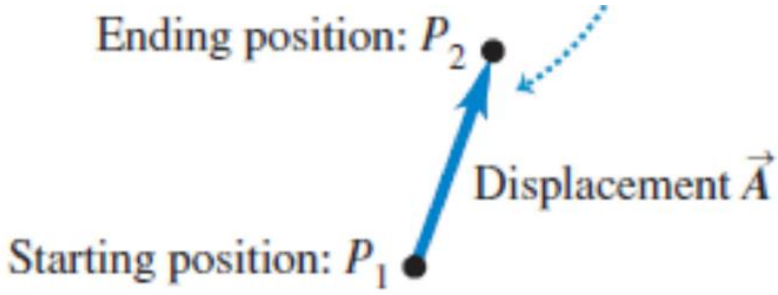
Scalar 标量

- Physical quantities can be scalar or vector.
- Scalar is a physical quantity that can be described by a number. It only has magnitude (“how much”, “how big”), but no direction.
- For examples:
  - Distance: 5m
  - Temperature: 25°C
  - Voltage: 100V



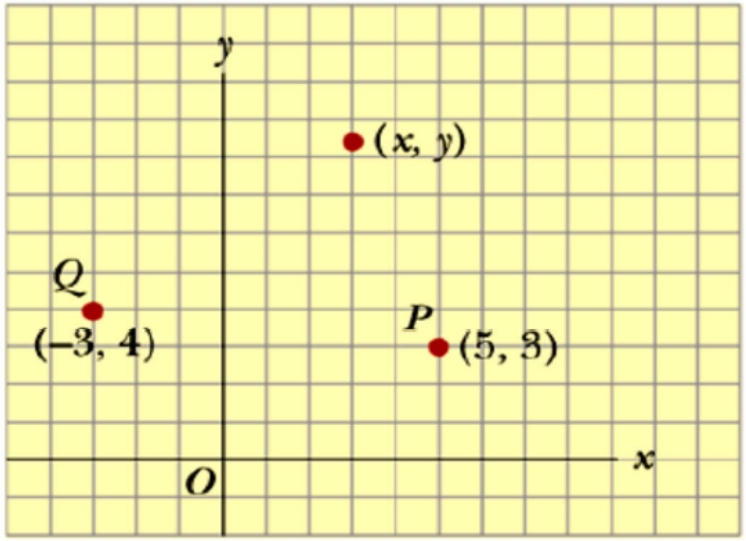
Vector 矢量

- Vector is a physical quantity that has both magnitude 大小 and direction in space.
- For examples: 方向
  - Displacement, **A**
  - Force, **F**

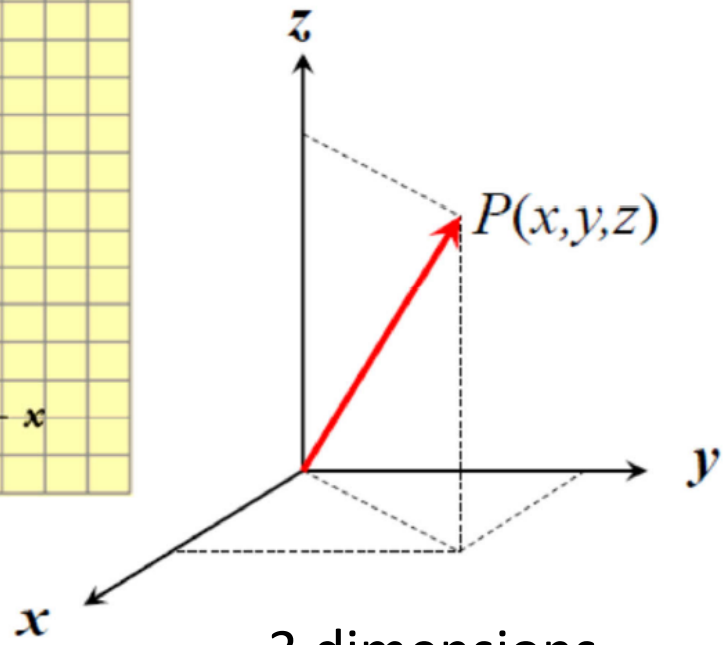


Representing Vector

- Vector is represented using Rectangular Coordinate System, also known as the Cartesian Coordinate System

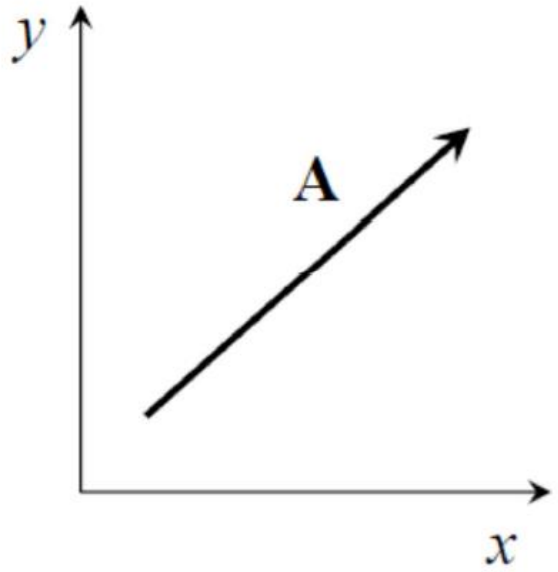


2 dimensions



3 dimensions

Representing Vector



Symbols:

A (bold face, printed materials)

$\vec{A}$ $\vec{A}$ (Italic with arrow, hand writing or printed materials)

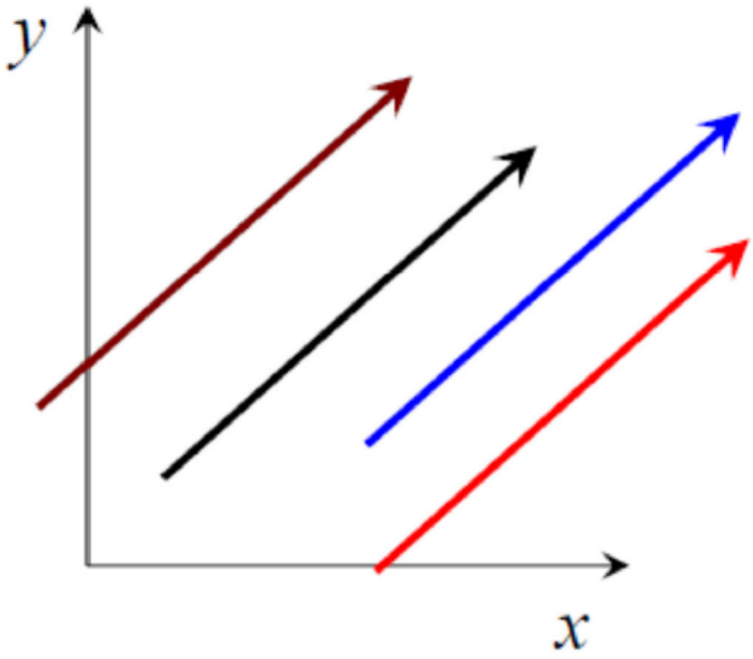
$$(\text{Magnitude of } \vec{A}) = A = |\vec{A}| \quad (1.1)$$

Magnitude = length of arrow

Direction = direction of arrow

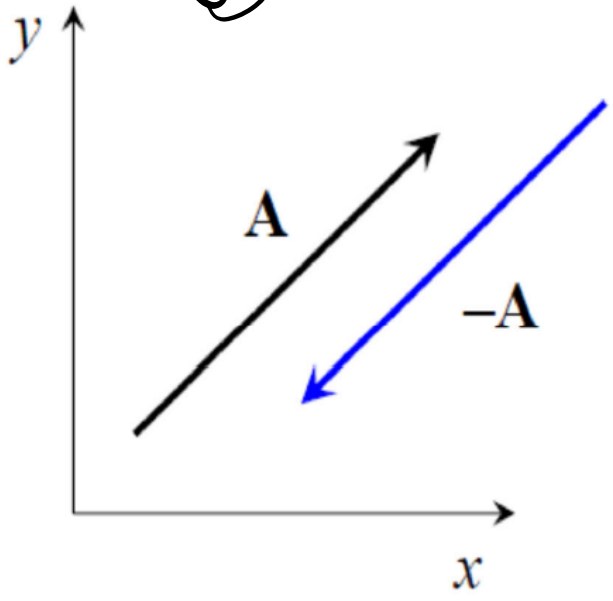
Properties of Vector

Equality of vectors



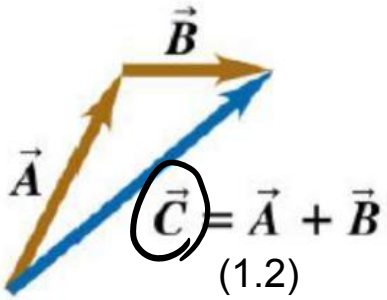
They must have same magnitude and same direction.

Negative of a vector

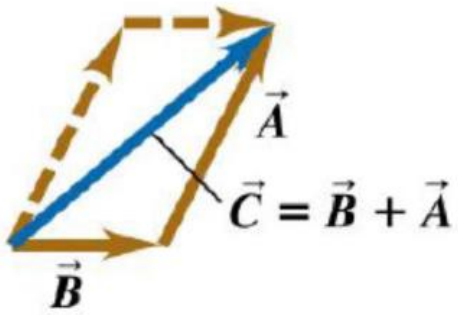


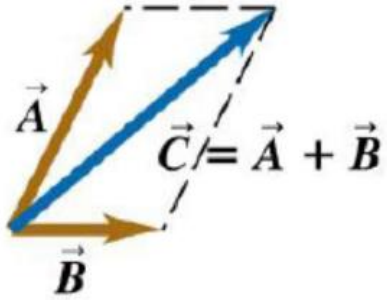
$$\mathbf{A} - \mathbf{A} = \mathbf{A} + (-\mathbf{A}) = \mathbf{0}$$

Addition for Vectors

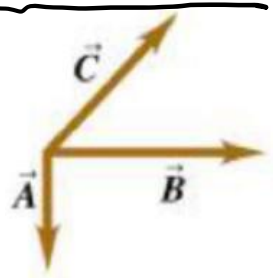


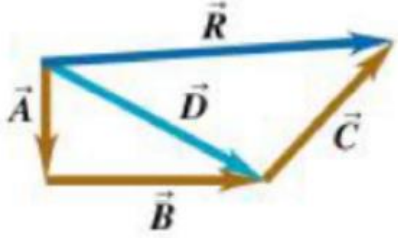
(1.2)

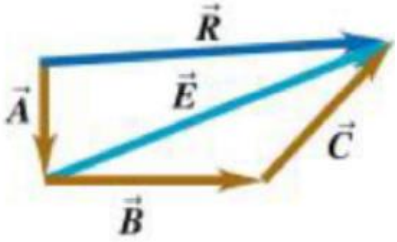




Commutative : $\vec{C} = \vec{B} + \vec{A}$ and $\vec{A} + \vec{B} = \vec{B} + \vec{A}$ (1.3)



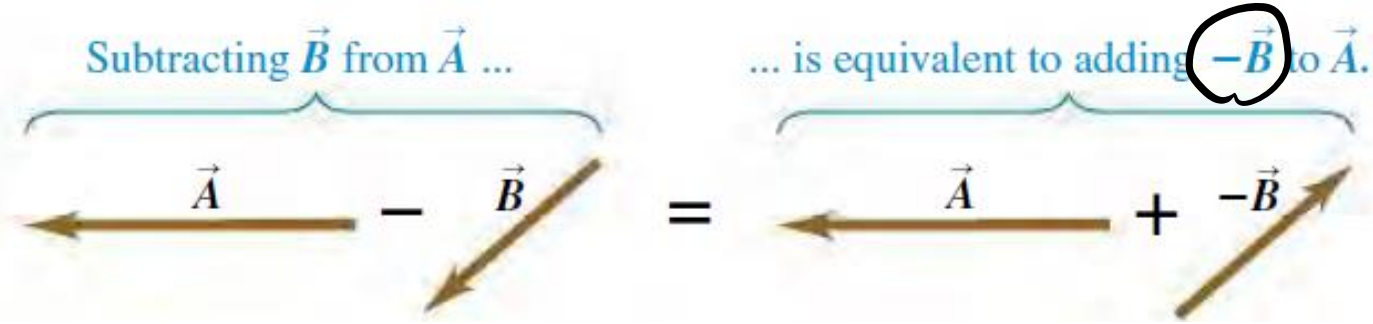




Associative: $\vec{R} = (\vec{A} + \vec{B}) + \vec{C} = \vec{D} + \vec{C}$

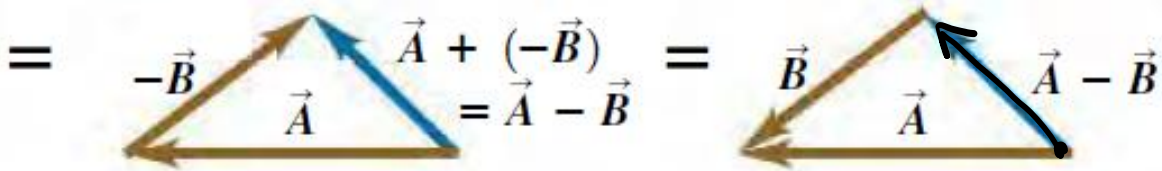
$\vec{R} = \vec{A} + (\vec{B} + \vec{C}) = \vec{A} + \vec{E}$

Subtraction for Vectors



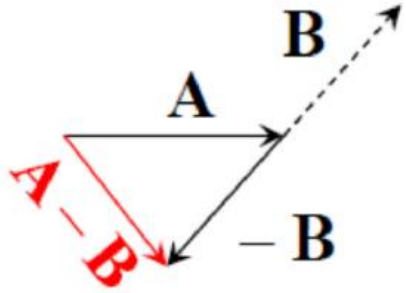
$$\vec{A} + (-\vec{B}) = \vec{A} - \vec{B}$$

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B}) \qquad (1.4)$$



With $\vec{A}$ and $-\vec{B}$ head to tail, $\vec{A} - \vec{B}$ is the vector from the tail of $\vec{A}$ to the head of $-\vec{B}$.

With $\vec{A}$ and $\vec{B}$ head to head, $\vec{A} - \vec{B}$ is the vector from the tail of $\vec{A}$ to the tail of $\vec{B}$.



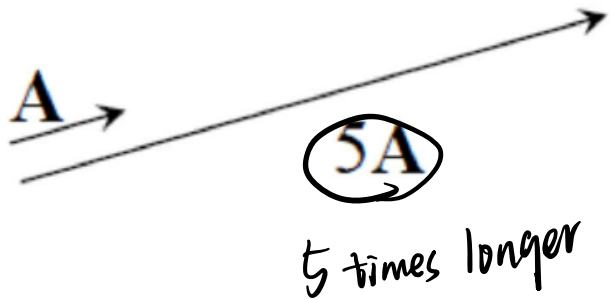
Multiplying a Vector by a Scalar

- If we multiply a vector $\vec{A}$ by a scalar *c*, each component of the product $\vec{D} = c\vec{A}$ is the product of *c* and the corresponding component of $\vec{A}$:

$$D_x = cA_x, \qquad D_y = cA_y \tag{1.8}$$

- Product of a Vector that has the same direction as $\vec{A}$ and magnitude *m* times of *A.*

If $m = 5$



Magnitude and Direction from Components of Vector

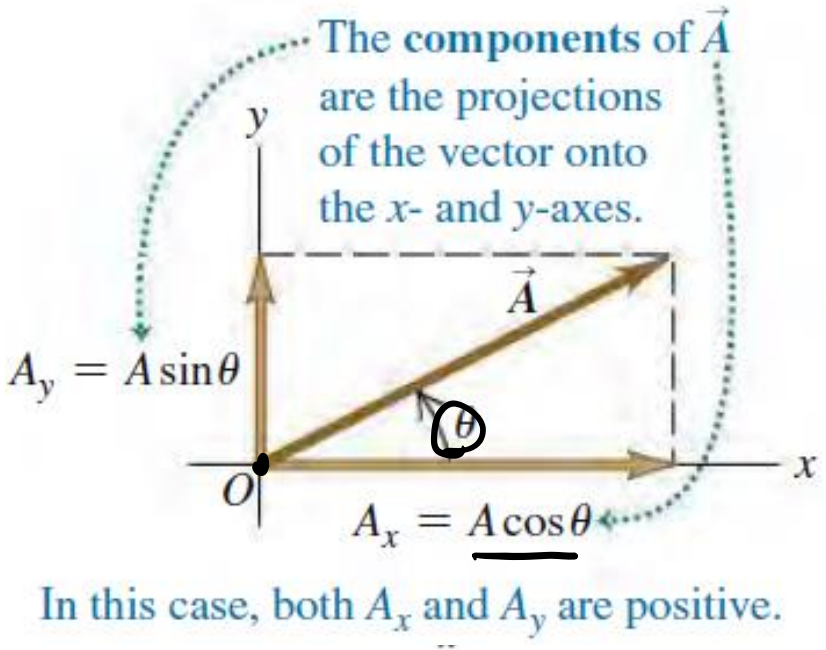
- Components are projections of a vector along the coordinate axes.

Magnitude

$$A_x = A\cos\theta$$
$$A_y = A\sin\theta \quad (1.5)$$

$$A = \sqrt{A_x^2 + A_y^2} \quad (1.6)$$

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right) \quad (1.7)$$



- Any vector can be described by its components.

Vector Sum (Resultant) with Components

- We can express the vector sum $\vec{R}$ of two 2D vectors as follows:
- We can use the same procedure to find the sum of any number of vectors.

$$R_x = A_x + B_x + C_x + D_x + E_x + \cdots$$
$$R_y = A_y + B_y + C_y + D_y + E_y + \cdots \quad (1.10)$$

- For 3D vector, third component:

$$R_z = A_z + B_z + C_z + D_z + E_z + \cdots$$

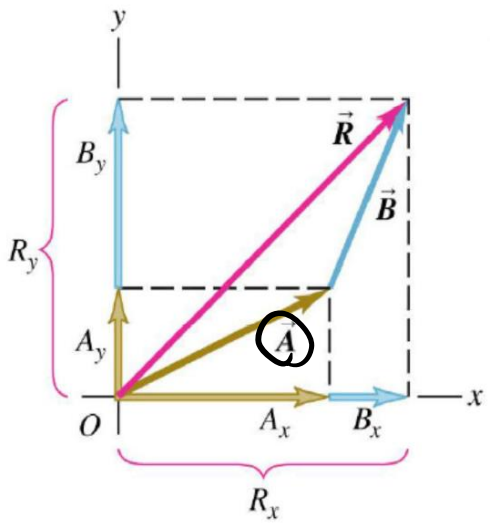
- Magnitude of 3D vector:

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad (1.11)$$

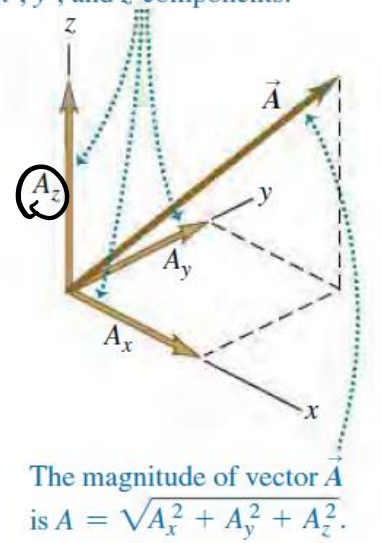
Each component of $\vec{R} = \vec{A} + \vec{B}$ ...

$$R_x = A_x + B_x, \qquad R_y = A_y + B_y \quad (1.9)$$

... is the sum of the corresponding components of $\vec{A}$ and $\vec{B}$.



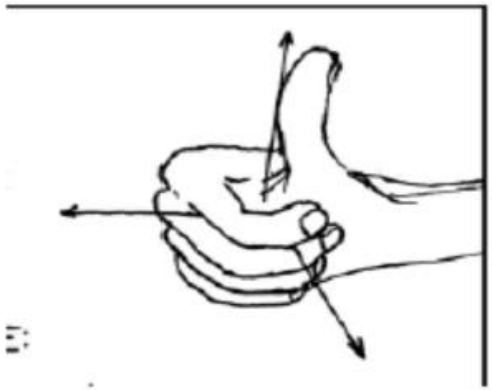
In three dimensions, a vector has x-, y-, and z-components.

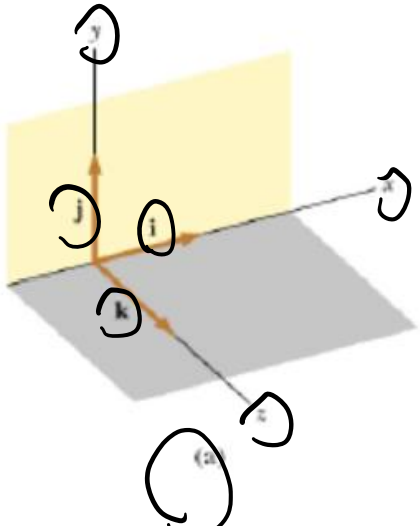


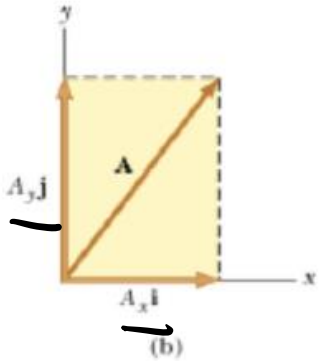
The magnitude of vector $\vec{A}$ is $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$.

Unit Vectors

- A dimensionless vector having a magnitude of exactly one.
- Particularly useful in a Cartesian coordinate system to introduce unit vectors *i, j, k* pointing in the positive x-, y-, and z-directions, respectively.
- Magnitude: $|i| = |j| = |k| = 1$
- *i, j, k* form a set of mutually perpendicular vectors in a right-handed coordinate system.

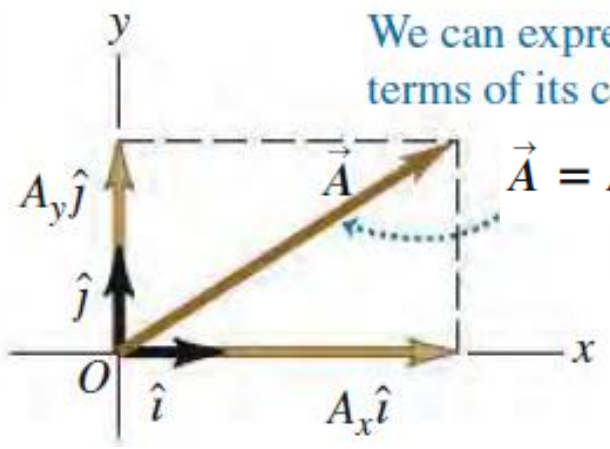






Unit Vectors

- Vector A in an xy-coordinate system can be written as:



We can express a vector $\vec{A}$ in terms of its components as

$$\vec{A} = A_x\hat{i} + A_y\hat{j} \qquad (1.12)$$

- The vector sum $\vec{R}$ of two vectors as follows:

$$\vec{A} = A_x\hat{i} + A_y\hat{j}$$

$$\vec{B} = B_x\hat{i} + B_y\hat{j}$$

$$\vec{R} = \vec{A} + \vec{B} \qquad (1.13)$$

$$= (A_x\hat{i} + A_y\hat{j}) + (B_x\hat{i} + B_y\hat{j})$$

$$= (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j}$$

$$= R_x\hat{i} + R_y\hat{j}$$

Unit Vectors

- Similarly with vector sum $\vec{R}$ of two 3D vectors:

Any vector can be expressed in terms of its x-, y-, and z-components ...

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$$

$$\vec{B} = B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}$$

... and unit vectors $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$.

(1.14)

$$\vec{R} = (A_x + B_x)\hat{\imath} + (A_y + B_y)\hat{\jmath} + (A_z + B_z)\hat{k}$$

$$= R_x\hat{\imath} + R_y\hat{\jmath} + R_z\hat{k}$$

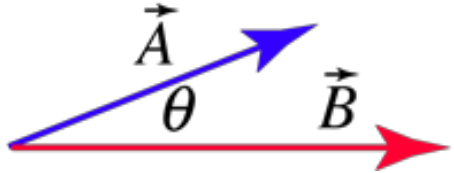
(1.15)

It is often easier to solve a problem by working with the vector components, and then finding the resultant vector (than doing complex geometry).

Scalar Product

- Also known as the "dot product".
- Although $\vec{A}$ and $\vec{B}$ are vectors, the quantity $\vec{A} \cdot \vec{B}$ is a scalar.
- The scalar product of two vectors can be expressed as:

$$\vec{A} \cdot \vec{B} = AB\cos\theta \qquad (1.16)$$



$\vec{A}$ denotes vector

A denotes the magnitude of the vector

- In rectangular coordinate system, since *i, j,* and *k* are unit vector with magnitude of 1 and they are perpendicularly with each other, we have

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = (1)(1)\cos 0^\circ = 1 \qquad (1.17)$$

$$\hat{i} \cdot \hat{j} = \hat{i} \cdot \hat{k} = \hat{j} \cdot \hat{k} = (1)(1)\cos 90^\circ = 0$$

Scalar Product

- Now we express $\vec{A}$ and $\vec{B}$ in terms of their components, expand the product, and use these products of unit vectors:

$$\vec{A}\cdot\vec{B} = (A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k})\cdot(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k})$$

$$= A_x\hat{\imath}\cdot B_x\hat{\imath} + A_x\hat{\imath}\cdot B_y\hat{\jmath} + A_x\hat{\imath}\cdot B_z\hat{k}$$

$$+ A_y\hat{\jmath}\cdot B_x\hat{\imath} + A_y\hat{\jmath}\cdot B_y\hat{\jmath} + A_y\hat{\jmath}\cdot B_z\hat{k}$$

$$+ A_z\hat{k}\cdot B_x\hat{\imath} + A_z\hat{k}\cdot B_y\hat{\jmath} + A_z\hat{k}\cdot B_z\hat{k}$$

$$= A_xB_x\hat{\imath}\cdot\hat{\imath} + A_xB_y\hat{\imath}\cdot\hat{\jmath} + A_xB_z\hat{\imath}\cdot\hat{k}$$

$$+ A_yB_x\hat{\jmath}\cdot\hat{\imath} + A_yB_y\hat{\jmath}\cdot\hat{\jmath} + A_yB_z\hat{\jmath}\cdot\hat{k}$$

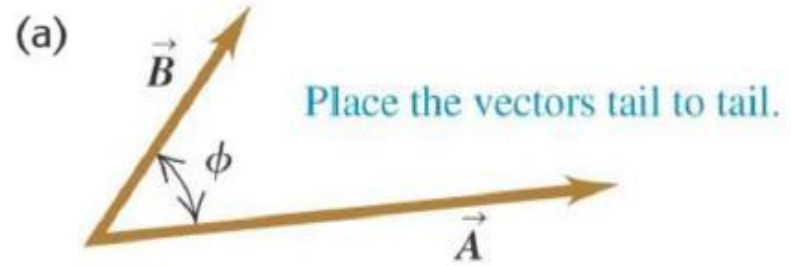
$$+ A_zB_x\hat{k}\cdot\hat{\imath} + A_zB_y\hat{k}\cdot\hat{\jmath} + A_zB_z\hat{k}\cdot\hat{k}$$

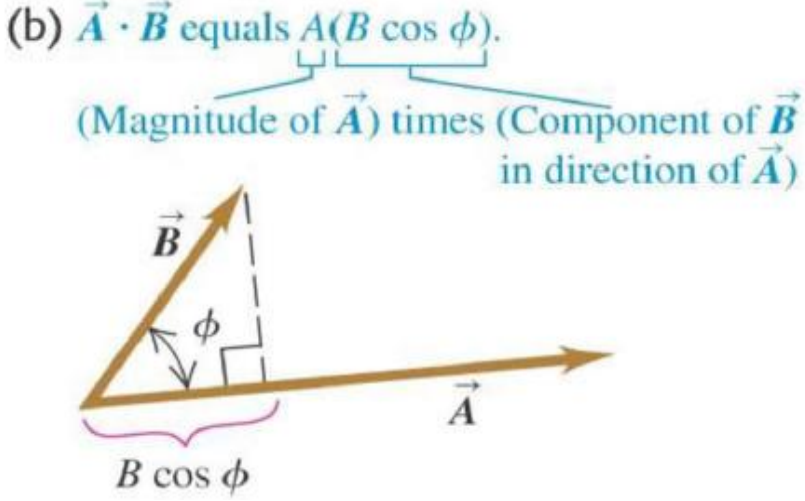
- Therefore,

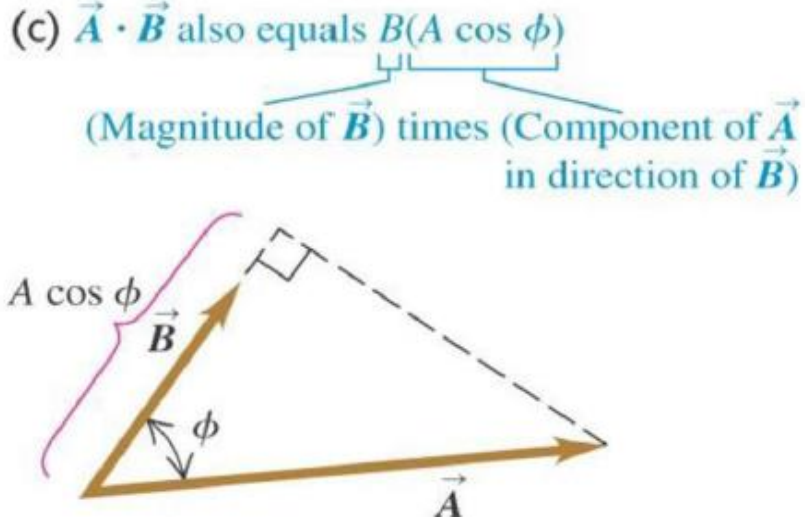
$$\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z \qquad (1.19)$$

Scalar Product

- Commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$

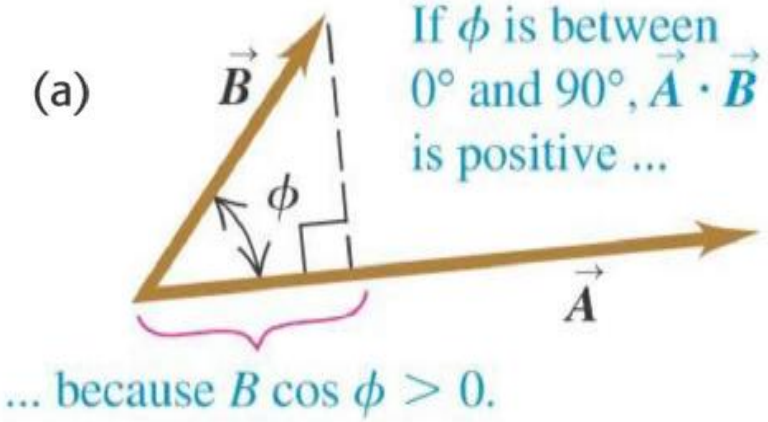


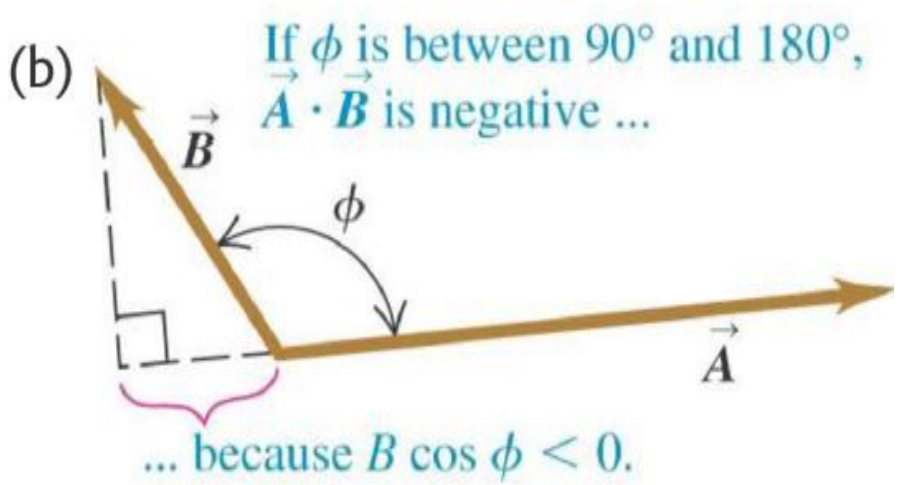


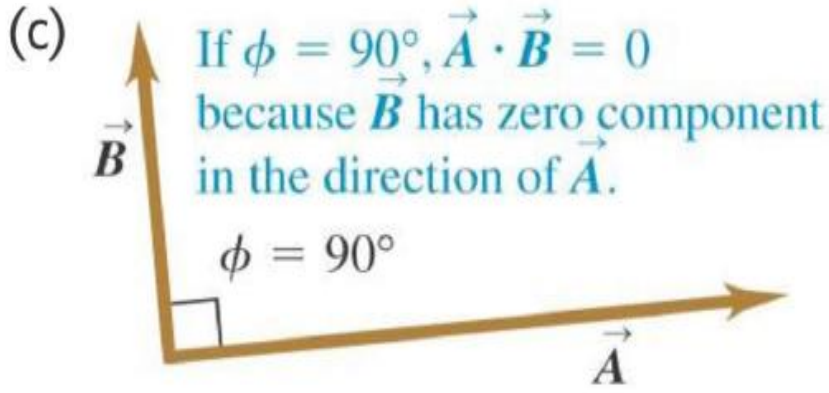


Scalar Product

- $\vec{A} \cdot \vec{B}$ is a scalar (or a number).
- It has no direction.
- It can be positive, negative, or zero.







Scalar Product

Example 1.10: Find the angle between the vectors

$$\vec{A} = 2.00\hat{i} + 3.00\hat{j} + 1.00\hat{k}$$
$$\vec{B} = -4.00\hat{i} + 2.00\hat{j} - 1.00\hat{k}$$

Solution: We know $\vec{A} \cdot \vec{B} = AB\cos\phi$ and $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

Then, $\cos\phi = \dfrac{\vec{A} \cdot \vec{B}}{AB} = \dfrac{A_x B_x + A_y B_y + A_z B_z}{AB}$

We can use this formula to find the angle between *any* two vectors $\vec{A}$ and $\vec{B}$. Here we have $A_x = 2.00$, $A_y = 3.00$, and $A_z = 1.00$, and $B_x = -4.00$, $B_y = 2.00$, and $B_z = -1.00$. Thus

$$\begin{aligned}\vec{A} \cdot \vec{B} &= A_x B_x + A_y B_y + A_z B_z \\ &= (2.00)(-4.00) + (3.00)(2.00) + (1.00)(-1.00) \\ &= -3.00\end{aligned}$$

Scalar Product

(con't)

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(2.00)^2 + (3.00)^2 + (1.00)^2}$$

$$= \sqrt{14.00}$$

$$B = \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(-4.00)^2 + (2.00)^2 + (-1.00)^2}$$

$$= \sqrt{21.00}$$

$$\cos\phi = \frac{A_xB_x + A_yB_y + A_zB_z}{AB} = \frac{-3.00}{\sqrt{14.00}\sqrt{21.00}} = -0.175$$

$$\phi = 100^\circ$$

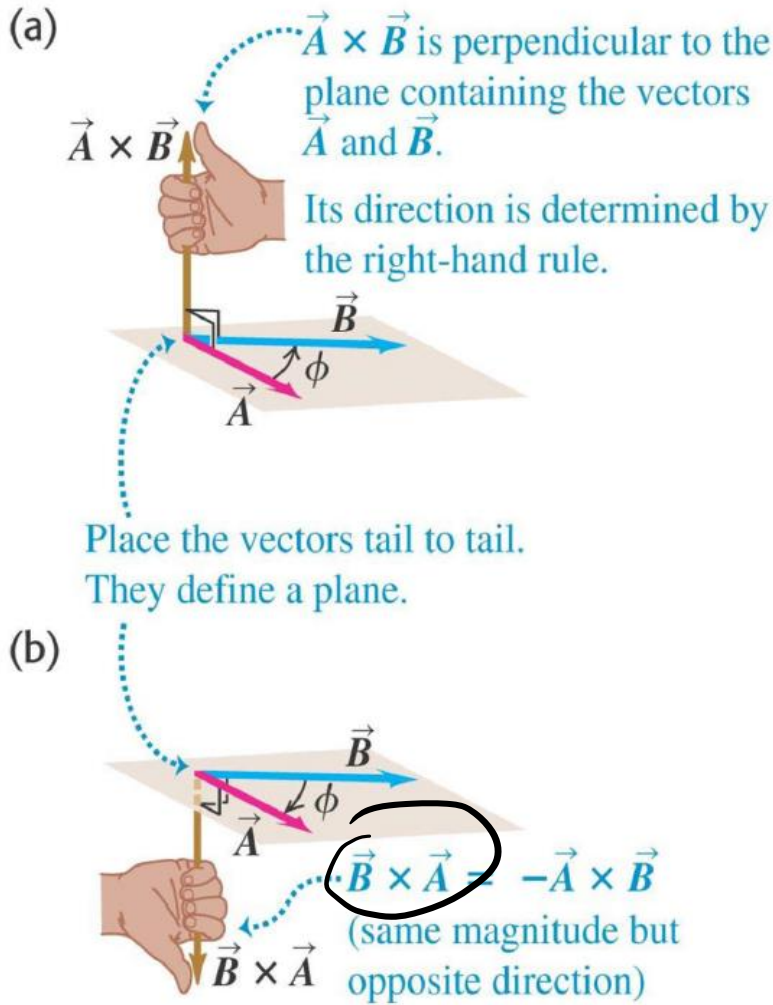
Vector Product

- Also known as the “cross product”.
- The vector product is itself a vector.
- If $\vec{C} = \vec{A} \times \vec{B}$, then its magnitude C is

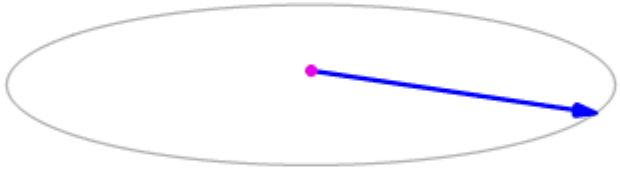
$$C = AB \sin\phi \quad \text{or} \quad C = \left|\vec{A}\times\vec{B}\right| = \left|\vec{A}\right|\left|\vec{B}\right| \sin\phi \tag{1.20}$$

- Direction of the vector product is determined by the right-hand rule.
- Anticommutative

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A} \tag{1.21}$$



Vector Product



Vector Product

- In rectangular coordinate system, since *i, j,* and *k* are unit vector with magnitude of 1 and they are perpendicularly with each other, we have

$$\hat{\imath} \times \hat{\imath} = \hat{\jmath} \times \hat{\jmath} = \hat{k} \times \hat{k} = \mathbf{0}$$

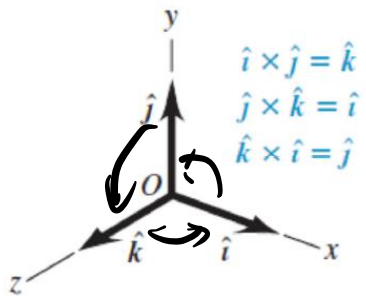
and

$$\hat{\imath} \times \hat{\jmath} = -\hat{\jmath} \times \hat{\imath} = \hat{k}$$

$$\hat{\jmath} \times \hat{k} = -\hat{k} \times \hat{\jmath} = \hat{\imath} \qquad (1.22)$$

$$\hat{k} \times \hat{\imath} = -\hat{\imath} \times \hat{k} = \hat{\jmath}$$

A right-handed coordinate system



- Expanding them

$$\begin{aligned}\vec{A} \times \vec{B} = (A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}) \times (B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}) \\ = A_x\hat{\imath} \times B_x\hat{\imath} + A_x\hat{\imath} \times B_y\hat{\jmath} + A_x\hat{\imath} \times B_z\hat{k} \\ + A_y\hat{\jmath} \times B_x\hat{\imath} + A_y\hat{\jmath} \times B_y\hat{\jmath} + A_y\hat{\jmath} \times B_z\hat{k} \\ + A_z\hat{k} \times B_x\hat{\imath} + A_z\hat{k} \times B_y\hat{\jmath} + A_z\hat{k} \times B_z\hat{k}\end{aligned} \qquad (1.23)$$

- Finally

$$\vec{A} \times \vec{B} = (A_yB_z - A_zB_y)\hat{\imath} + (A_zB_x - A_xB_z)\hat{\jmath} + (A_xB_y - A_yB_x)\hat{k} \qquad (1.24)$$

Components of vector (cross) product $\vec{A} \times \vec{B}$

$$C_x = A_yB_z - A_zB_y \qquad C_y = A_zB_x - A_xB_z \qquad C_z = A_xB_y - A_yB_x$$

A_x, A_y, A_z = components of $\vec{A}$ $\qquad$ B_x, B_y, B_z = components of $\vec{B}$ $\qquad$ (1.25)

Vector Product

- Alternatively, the vector product can also be compactly stated in the form of a determinant which for the 3x3 case has a convenient evaluation procedure:

$$\vec{A}\times\vec{B}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}$$

$$\begin{vmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{vmatrix} = a_{11}\left(a_{22}a_{33}-a_{23}a_{32}\right)-a_{12}\left(a_{21}a_{33}-a_{23}a_{31}\right)+a_{13}\left(a_{21}a_{32}-a_{22}a_{31}\right)$$

First row element; Cofactor; Alternating signs; First row element; Cofactor; etc.

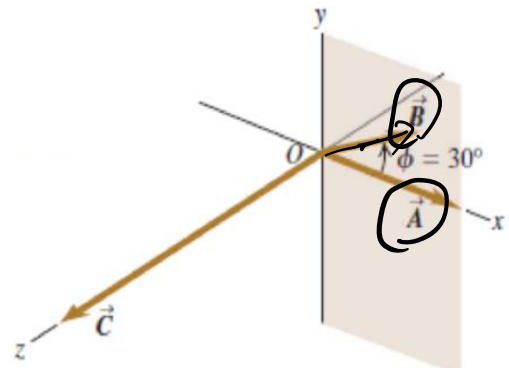
- Once the scheme for determinant evaluation is familiar, this is a convenient way to reconstruct the expanded form:

$$\vec{A}\times\vec{B}=\left(A_yB_z-A_zB_y\right)\hat{i}+\left(A_zB_x-A_xB_z\right)\hat{j}+\left(A_xB_y-A_yB_x\right)\hat{k} \qquad (1.24)$$

Vector Product

EXAMPLE 1.11 CALCULATING A VECTOR PRODUCT

Vector $\vec{A}$ has magnitude 6 units and is in the direction of the $+x$-axis. Vector $\vec{B}$ has magnitude 4 units and lies in the xy-plane, making an angle of 30° with the $+x$-axis (**Fig. 1.33**). Find the vector product $\vec{C} = \vec{A} \times \vec{B}$.



1.33 Vectors $\vec{A}$ and $\vec{B}$ and their vector product $\vec{C} = \vec{A} \times \vec{B}$. Vector $\vec{B}$ lies in the xy-plane.

Solution: since

$$\left|\vec{A}\times\vec{B}\right| = \left|\vec{A}\right|\left|\vec{B}\right|\sin\phi = AB\sin\phi$$

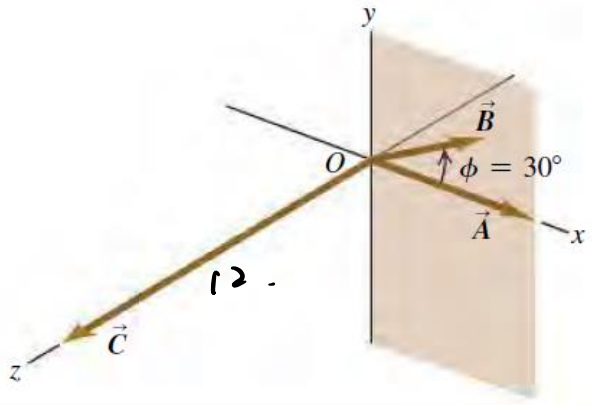
and

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

we can use these two equations to solve the problem.

Vector Product

(cont'd)



From Eq. (1.20) the magnitude of the vector product is

$$AB \sin \phi = (6)(4)(\sin 30°) = 12$$

By the right-hand rule, the direction of $\vec{A} \times \vec{B}$ is along the $+z$-axis (the direction of the unit vector $\hat{k}$), so $\vec{C} = \vec{A} \times \vec{B} = 12\hat{k}$.

We can also expand $\vec{A} \times \vec{B}$ to get Eq (1.24), but we first determine the components of vector **A** and **B**:

$$A_x = 6 \qquad A_y = 0 \qquad A_z = 0$$

$$B_x = 4 \cos 30° = 2\sqrt{3} \qquad B_y = 4 \sin 30° = 2 \qquad B_z = 0$$

Vector Product

(cont'd)

After obtaining the expanded form:

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{\imath} + (A_z B_x - A_x B_z)\hat{\jmath} + (A_x B_y - A_y B_x)\hat{k}$$

Then Eqs. (1.25) yield

$$C_x = (0)(0) - (0)(2) = 0$$

$$C_y = (0)(2\sqrt{3}) - (6)(0) = 0$$

$$C_z = (6)(2) - (0)(2\sqrt{3}) = 12$$

Thus again we have $\vec{C} = 12\hat{k}$.

Evaluation: *Both methods give the same results. Depending the situation, one or the other of the two approaches may be the more convenient one to use.*

Summary

- Chapter 1 has been discussed
- Reading materials
  - Chapter 1 in the text book
- Assignment 1
  - Homework on Units, Physical Quantities, and Vector will be assigned soon, check Moodle announcement for updates